ADRIAN LEON SCHOLL

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RESEARCH INTERESTS

Methods: uncertainty quantification and Bayesian deep learning, in particular calibrated aleatoric and epistemic uncertainty, also in deep generative models; generative modeling with flow matching; probabilistic forecasting.
Applications: prior work in energy systems and renewables (electricity price and demand forecasting) and in armed-conflict prediction (German Federal Foreign Office); strong interest in medical AI.

EDUCATION

M.Sc. Data Science, Elite Graduate Program Oct 2023 – Aug 2026
Ludwig-Maximilians-Universität (LMU) Munich, Germany *Final grade: **1.09** (German scale, 1.0 = best)*

- Thesis: *Evaluating Adaptive Timestep Sampling for the Efficient Training of Flow Matching Models*, grade **1.0**, defense **1.0**. Supervisor: Prof. Dr. Björn Ommer; advisor: Felix Krause (Computer Vision & Learning Group)
- Selected courses: Inference and Sampling, two-semester core module incl. Statistical Reasoning and Inference (Bayesian inference, MCMC; 12 ECTS, oral exam, 1.0); Advanced Statistical Modelling and Programming (1.0); Multivariate Statistics and Database Management (1.0)
- Seminar paper and talk (**1.0**): DiBS, Bayesian causal structure learning with SVGD (Lorch, Rothfuss, Schölkopf, Krause; NeurIPS 2021), own experiments on graph priors and posterior variance. [code]

Erasmus exchange, Master's in Statistics and Operations Research Feb 2025 – Jul 2025
Universitat Politècnica de Catalunya (UPC), Barcelona, Spain

- Courses: Statistical Learning with Deep Artificial Neural Networks (Honors) and Time Series (9.5/10), both recognized at LMU with 1.0
- Time series project (with L. Bergstrand): SARIMA and ARIMAX models of Spain's monthly electricity demand, 1990–2019; calendar and outlier effects cut the 2019 hold-out MAPE from 2.3% to 1.7% [report] [code]

B.Sc. Industrial Engineering and Management Oct 2018 – Nov 2022
Karlsruhe Institute of Technology (KIT), Germany *Final grade: **1.3***

- Thesis: *Development of an Automated Conflict Prediction System by State Space ARIMA Methods*, grade **1.0**, in cooperation with the Federal Foreign Office. Supervisor: Prof. Dr. Melanie Schienle; advisor: Lotta Rüter
- Selected courses, all 1.0: Introduction to Machine Learning, Data Mining and Applications, Statistics I and II, Introduction to Econometrics, Financial Data Science, Operations Research I and II, Database Systems

Abitur (university entrance qualification) Sep 2009 – Jul 2017
Leibniz-Gymnasium Östringen, Germany *Final grade: **1.1***

RESEARCH EXPERIENCE

Master's Thesis Research Nov 2025 – Aug 2026
Computer Vision & Learning Group (Prof. Dr. Björn Ommer), LMU Munich *Munich, Germany*

- Trained reinforcement-learning policies (REINFORCE, A2C and a curvature-aware variant) to choose the training timesteps of conditional optimal-transport flow matching models on CIFAR-10
- Showed that the adaptive schedules do not improve on uniform sampling and explained why through per-timestep loss and gradient-variance dynamics and an analysis of policy-gradient behavior
- Built and ran the multi-GPU training pipeline independently (PyTorch DDP, SLURM, W&B, 8-bit optimizers)

Data Science Practical with forrs (industry partner) Oct 2024 – Feb 2025
Department of Statistics (Prof. Dr. Göran Kauermann), LMU Munich *Munich, Germany*

- Built an hourly price forward curve for the German power market (2–5 year horizon) in a team of two
- Combined a Bayesian seasonal model with uncertainty intervals with an arbitrage-free quartic-spline correction that keeps the curve consistent with traded futures; grade **1.0**

Data Science Consultant (self-employed) and Intern Oct 2021 – Sep 2022
German Federal Foreign Office, S 05 PREVIEW Crisis Early Warning *Berlin, Germany*

- Built an automated state-space ARIMA framework (B.Sc. thesis) that beat the no-change baseline in one-year-ahead forecasts of conflict fatalities on all metrics and improved on Auto-ARIMA by 56% in MAE. [code]
- Initiated a long-term research cooperation on conflict prediction between the Federal Foreign Office and the group of Prof. Dr. Melanie Schienle at KIT; presented forecasts at internal expert workshops

THESES

- A. Scholl. *Evaluating Adaptive Timestep Sampling for the Efficient Training of Flow Matching Models.* Master's thesis, LMU Munich, 2026. adrian-io.github.io/thesis.html

- A. Scholl. *Development of an Automated Conflict Prediction System by State Space ARIMA Methods.* Bachelor's thesis, Karlsruhe Institute of Technology, 2022.

TALKS AND PRESENTATIONS

- *Bayesian uncertainty quantification for energy price forecasting.* Workshop (co-taught), international summer school "Meteorology and Markets", Hamburg, Jun 30 – Jul 4, 2025. Invited by Prof. Dr. Göran Kauermann
- *Evaluating adaptive timestep sampling for flow matching.* Master's thesis defense, LMU Munich, 2026
- *Differentiable Bayesian causal structure learning.* Seminar talk, LMU Munich, Jan 2025
- *Hourly price forward curve for the German power market.* Final project presentation, Data Science Practical with forrs, LMU Munich, 2025
- *Machine-learning prediction of protests, armed conflict and security-relevant incidents in the Middle East.* Internal expert workshops, German Federal Foreign Office, Berlin, 2021–22

TEACHING

- Co-instructor, summer-school workshop on uncertainty quantification in energy price forecasting, Hamburg 2025
- Tutor, Economics III: Introduction to Econometrics, KIT: exercise sessions in R Apr – Sep 2021
- Tutor, Statistics II, KIT: exercise sessions for about 50 students in two groups Nov 2020 – Mar 2021
- Tutor, preparatory course in object-oriented programming (Java), MINT-Kolleg, KIT Sep 2019

SCHOLARSHIPS AND AWARDS

- Travel grant, international summer school "Meteorology and Markets", Hamburg 2025
- Erasmus+ scholarship, exchange at UPC Barcelona 2025
- Admission to the Elite Graduate Program Data Science (Elite Network of Bavaria), LMU Munich 2023
- Deutschlandstipendium (Germany Scholarship), sponsored by QuantCo, KIT 2020 – 2021

INDUSTRY EXPERIENCE

Working Student Machine Learning Nov 2024 – Jan 2025
Scalable Capital, Data Science Unit *Munich, Germany*

- Built an automated evaluation framework for retrieval-augmented generation (RAG) in AI chatbot agents: HyDE retrieval, chunking and ranking strategies, LLM-as-a-judge (LlamaIndex), parallel asynchronous evaluation

Intern, then Working Student, Data Analytics & Analytics Engineering Apr 2023 – Sep 2024
FINN, Fleet Intelligence Unit *Munich, Germany*

- Developed a cash forecasting model for long-term liquidity planning and an ML model of vehicle underutilization costs; solely responsible for ∼400 data-warehouse models (dbt, BigQuery SQL); analyses for the Series C round

Pre-Sales Intern, Public Services and Smart Cities Jun 2018 – Sep 2018
SAP SE *Walldorf, Germany*

- Analyzed the smart-city start-up market and partnership opportunities; presented smart-city use cases of SAP IoT, S/4HANA Cloud and AI/ML solutions at the Smart City Forum

SELECTED PROJECTS

- **Aurora Media** (2026): AI consulting case at McKinsey / QuantumBlack's "New Horizon" event, Barcelona: demo of an agentic recommender (n8n) with semantic search and personalized outreach. [code]
- **EthXPose** (2025, course paper with J. Wankmüller): detection of fraudulent Ethereum wallets with transaction-graph embeddings and gradient boosting (93.9% accuracy), served as a FastAPI web service. [paper] [code]
- **VISTA** (2024): tracks study focus via 3D gaze vectors from webcam video (OpenCV, PyTorch CNN, FastAPI)
- **Commercial real estate default prediction** (2021, with L. Bleher, KIT course): logistic regression, SVMs and random forests with SMOTE for rare loan defaults. [code]

SKILLS

Languages	German (native), English (C1), Spanish (B2), French (A2)
Programming	Python, R, SQL, Java; Git, Docker, SLURM, FastAPI, Pydantic, SQLAlchemy
ML and statistics	PyTorch (DDP), scikit-learn, TensorFlow, OpenCV, LlamaIndex, statsmodels, sktime, SciPy, CVXPY
Data	BigQuery, dbt, Weights & Biases, Looker / LookML, Tableau

REFERENCES

- **Prof. Dr. Björn Ommer**, Head of the Computer Vision & Learning Group, Institute of Computer Science, LMU Munich; b.ommer@lmu.de
- **Prof. Dr. Göran Kauermann**, Chair of Applied Statistics in Social Sciences, Economics and Business, Department of Statistics, LMU Munich; goeran.kauermann@stat.uni-muenchen.de